

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455852.8781 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.117 +/- 0.016
Transit depth (Rp/Rs)^2	1.37 +/- 0.36 [%]
Orbital Inclination (inc)	85.03 +/- 1.36
Airmass coefficient 1 (a1)	1.358 +/- 0.011
Airmass coefficient 2 (a2)	-0.1281 +/- 0.0028
Scatter in the residuals of the lightcurve fit is	0.81 %
Best Comparison Star	None
Optimal Aperture	3.86
Optimal Annulus	7.69
Transit Duration (day)	0.1383 +/- 0.0086

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.117 $\pm$ 0.016 vs 0.1036 (deviation 0.66 σ)
Duration fit vs published	0.1383 d vs 0.1567 d (deviation 1.69 σ)
Mid-time O–C	6.6 min
Depth SNR	5.8
Residual scatter	0.81 %

Light curve

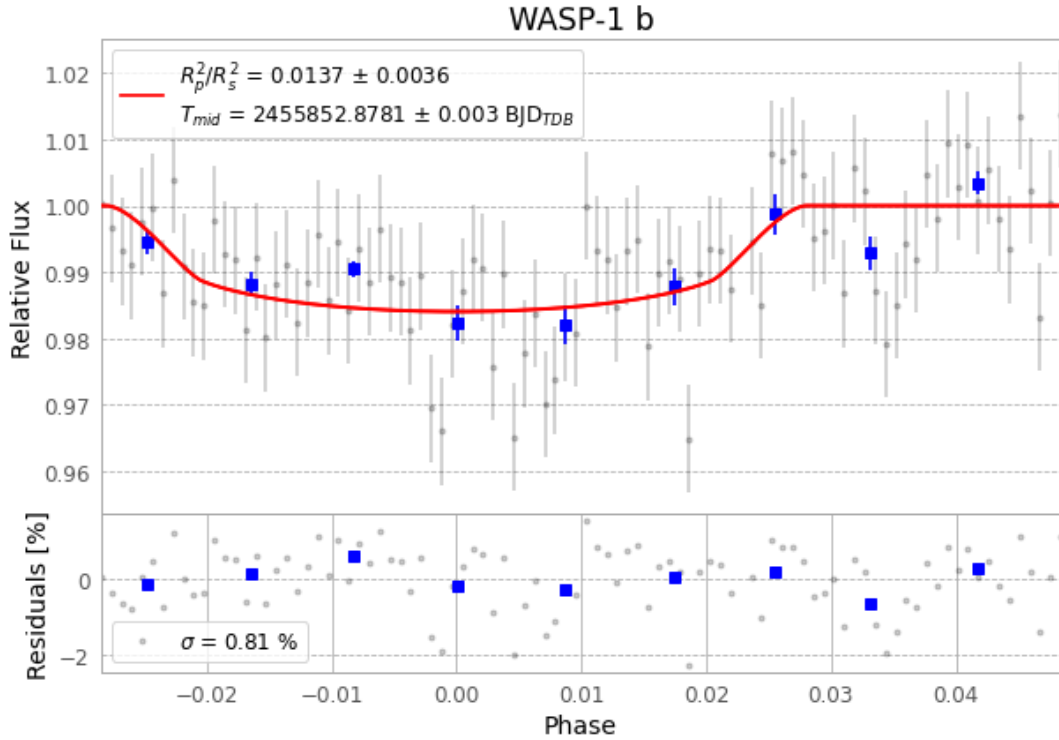


Figure 1 — FinalLightCurve_WASP-1 b_2011-10-18.png (SHA-256 f19525bf2b8aefe3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [242, 304], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c7912f22bc965e3c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

94 FITS frames (62 MB), WASP-1111018071212.FITS → WASP-1111018115112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-111018.log (9fcadb5bce2c...), FinalLightCurve_WASP-1 b_2011-10-18.png (f19525bf2b8a...), FinalLightCurve_WASP-1 b_2011-10-18.csv (e1bc8b7778b9...)

Compute resources

Wall time 128.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-15 23:35Z.